

OG-PHL: PEP Simulation Results

Jason DeBacker¹ **Richard W. Evans**²

¹University of South Carolina, Department of Economics

²Abundance Institute, Open Research Group, Inc.

January 16, 2026
United Nations, Philippines

PEP Simulation Overview

- The Philippines Energy Plan (PEP) 2023 simulation models the economic and health impacts of energy sector transitions
- We compare a baseline scenario to the PEP 2023 scenario over a 50-year time horizon
- Key dimensions analyzed:
 - Energy sector parametrization (TFP, costs, investments)
 - Environmental outcomes (CO2e, PM2.5)
 - Health impacts (mortality rates, deaths averted)
 - Macroeconomic effects (GDP, capital, consumption, labor)
 - Fiscal implications (government spending, debt)
 - Distributional impacts (prices, consumption, income, labor supply)

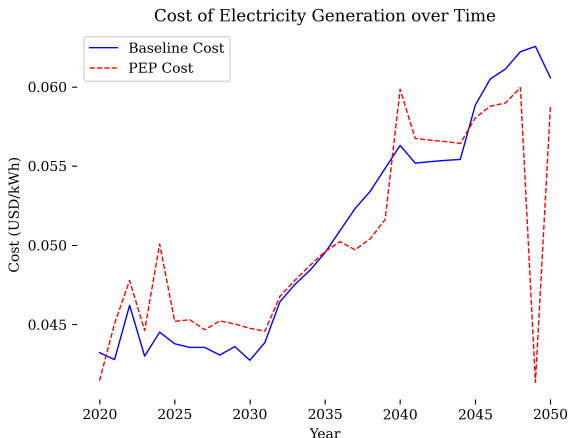
Input-Output Table for Multi-Industry Calibration

	Natural Resources	Electricity	Construction, Trade, Services	Manu.
Food	0.120	0.000	0.202	0.679
Energy & water	0.021	0.107	0.329	0.543
Non-durables	0.043	0.006	0.628	0.323
Durables	0.011	0.011	0.236	0.743
Services	0.041	0.003	0.772	0.185

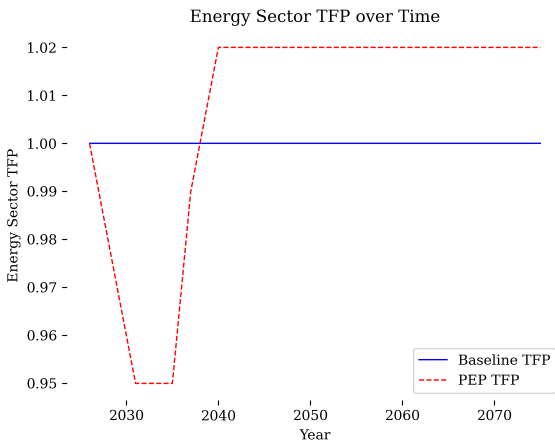
Consumption Category Shares for Multi-Industry Calibration

Consumption Category	Expenditure Share
Food	0.357
Energy and water	0.014
Non-durables	0.021
Durables	0.103
Services	0.505

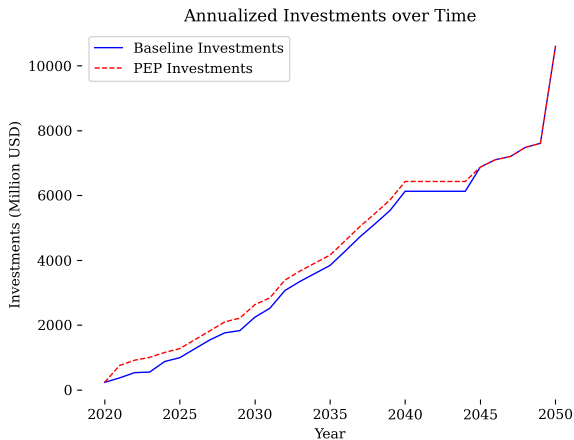
CLEWs: Cost of Electricity Generation over Time



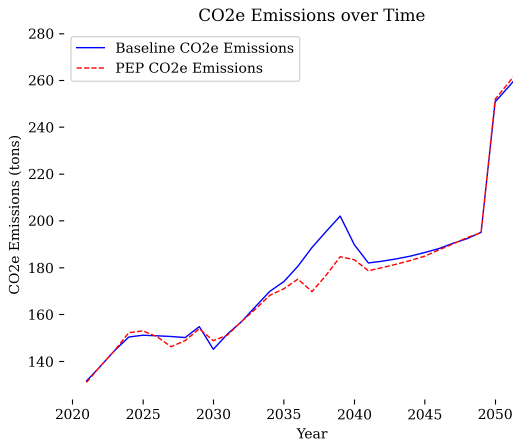
OG-PHL: Energy Sector TFP over Time



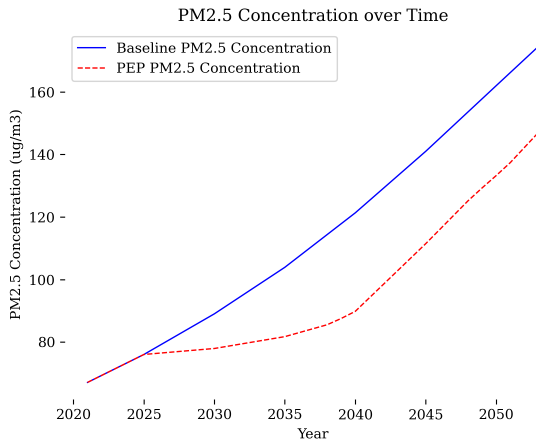
CLEWs: Annualized Investments over Time



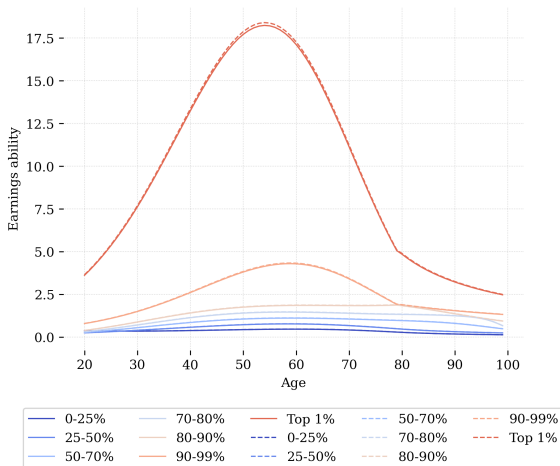
CLEWs: CO₂e Emissions over Time



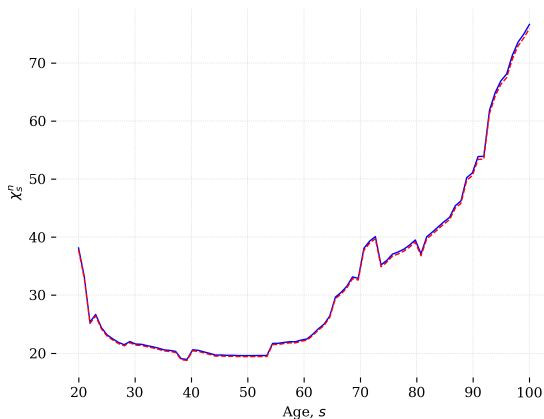
CLEWs: PM2.5 Concentration over Time



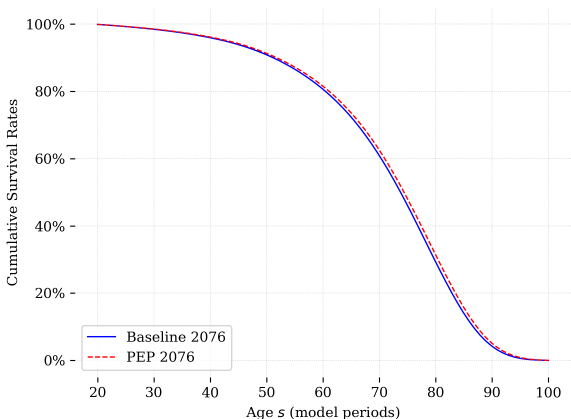
Ability Profiles in Baseline and PEP



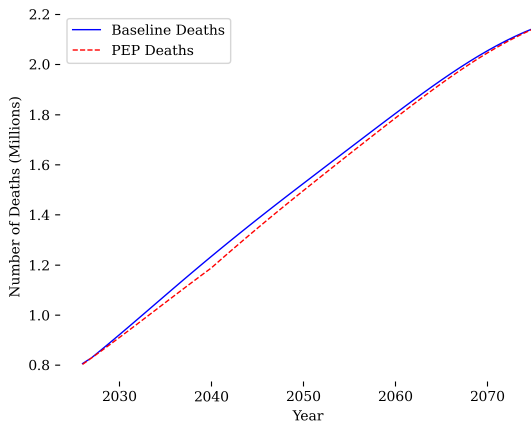
Disutility of Labor in Baseline and PEP



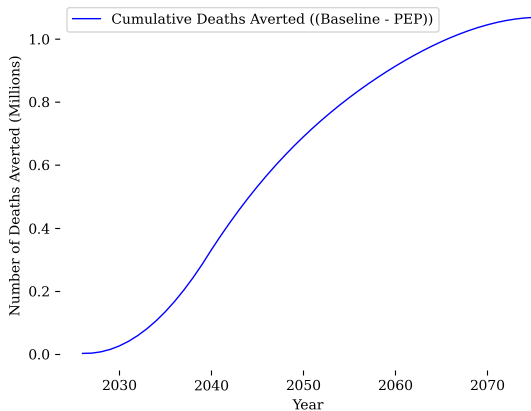
Mortality Baseline and PEP



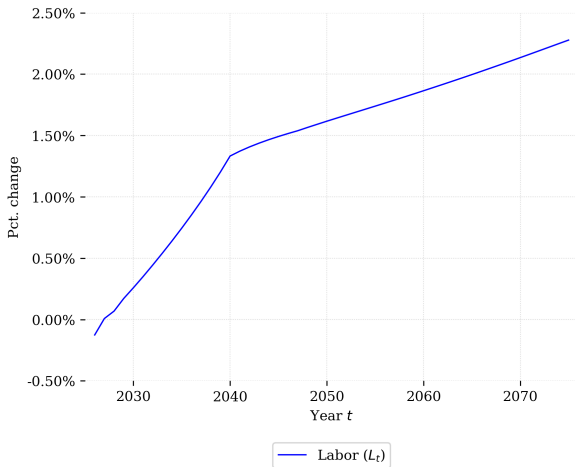
Annual Deaths over Time



Cumulative Deaths Averted over Time

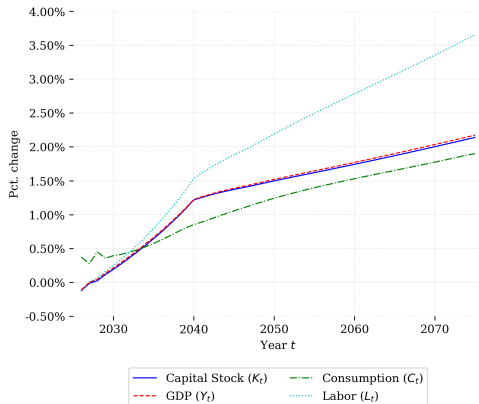


Labor Supply over Time



Aggregate Variables over Time

- Percentage changes in capital (K), output (Y), consumption (C), and labor (L)

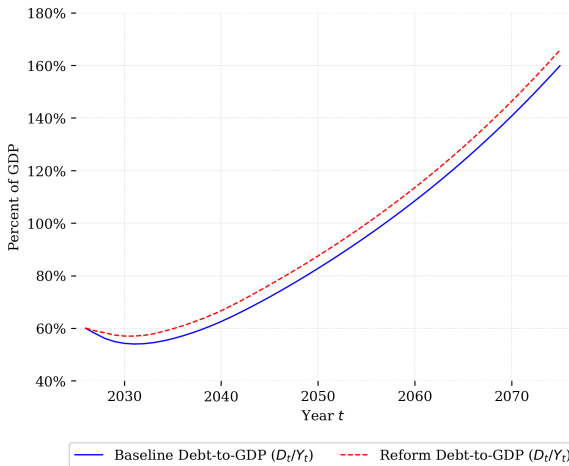


Long Run Effects on GDP

Discount Rate	NPV of GDP Increase (Trillions of PHP)
1%	5,073.030
2%	2,216.387
3%	998.952
4%	466.805
5%	227.508

Note: PEP23 Estimate of costs: 26-42 Trillion PHP

Debt-to-GDP Ratio over Time

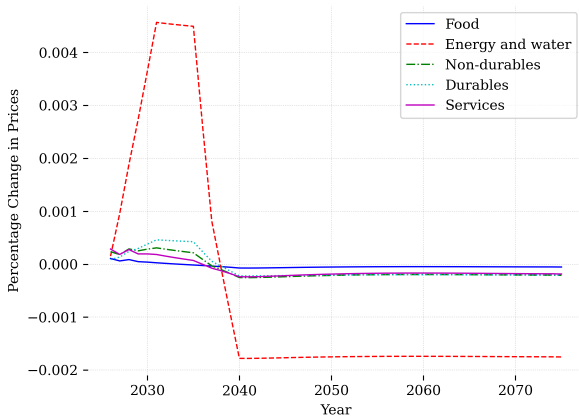


Long run revenue effects

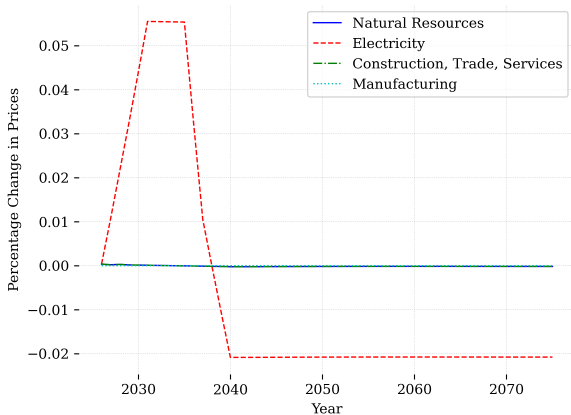
Discount Rate	NPV of Tax Revenue Increase (Trillions PHP)
1%	294.667
2%	131.213
3%	60.710
4%	29.408
5%	15.041

Note: PEP23 Estimate of government costs: 2.6-4.2 Trillion PHP

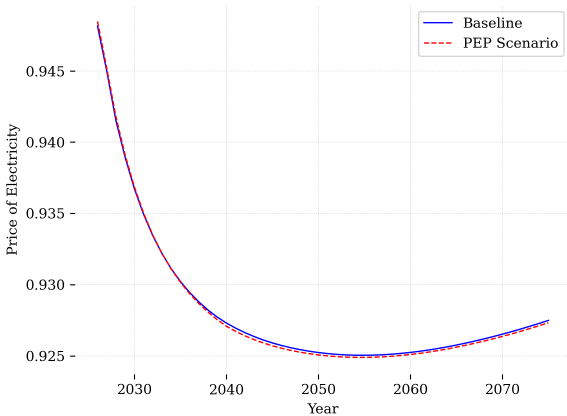
Percentage Change in Consumption Good Prices



Percentage Change in Production Good Prices

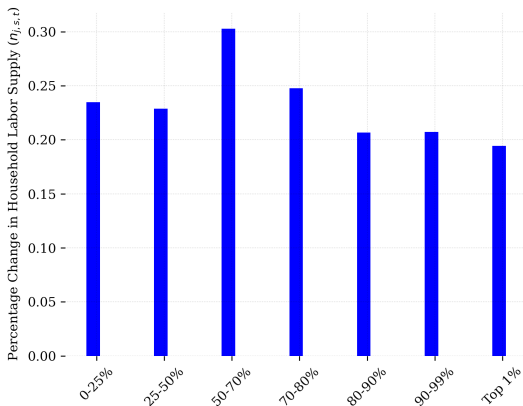


Price of Electricity



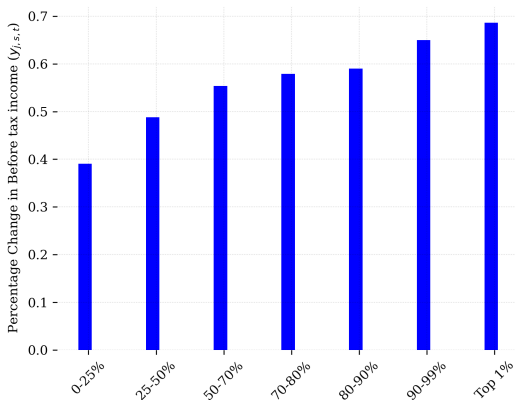
Percentage Change in Labor Supply by Ability Group

- Changes 10-20 years after policy implementation



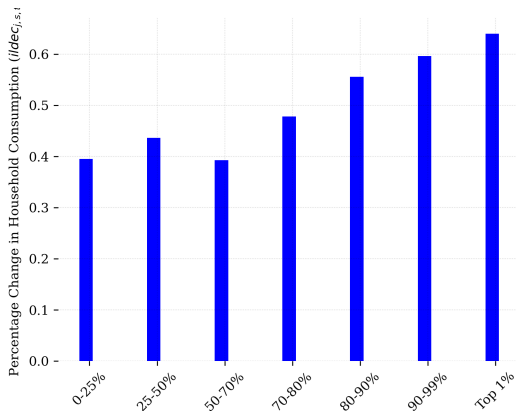
Percentage Change in Income by Ability Group

- Changes 10-20 years after policy implementation



Percentage Change in Consumption by Ability Group

- Changes 10-20 years after policy implementation



Key Findings

- Energy Transition:
 - PEP scenario shows increased energy sector TFP over time
 - Lower electricity generation costs in PEP scenario
 - Significant upfront investments required
- Environmental Benefits:
 - Reduced CO2e emissions
 - Lower PM2.5 concentrations
- Health Impacts:
 - Improved survival rates and lower mortality
 - Substantial cumulative deaths averted
 - Increased labor supply due to better health

Key Findings (continued)

- Macroeconomic Effects:
 - Positive impacts on capital, output, consumption, and labor
- Fiscal Implications:
 - Government spending increases for energy investments
 - Changes in debt-to-GDP ratio
- Distributional Effects:
 - Changes in relative prices across sectors
 - Electricity prices adjust over time
 - Differential impacts across ability groups